

Abdel-Rhman Adel Abdel-Samii

Electronics & Communication Engineering Student

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GitHub



LinkedIn



PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience designing and deploying cloud-connected embedded systems. Extensive self-driven training in AWS with practical expertise building serverless IoT architectures. Proficient in low-level embedded development (AVR, AUTOSAR). Eager to apply cloud engineering skills in enterprise environments.

EDUCATION

Misr International University (MIU) — Cairo, Egypt

Sep 2021 – Jul 2026

B.Sc. Electronics and Communication Engineering | GPA: 3.21 / 4.00

PROJECTS

Smart Campus Lighting System

AWS Lambda · DynamoDB · IoT Core · EventBridge · API Gateway · ESP8266 · Flutter · MQTT/TLS · Python

- Serverless, schedule-driven lighting automation integrating AWS cloud services with IoT hardware.
- EventBridge-triggered Lambda (Python 3.12) evaluating lecture schedules every minute, publishing control bit-masks via MQTT/TLS to AWS IoT Core.
- Two DynamoDB tables with composite keys: a permanent schedule store and a TTL-enabled temporary-session table for efficient per-device, per-day queries.
- REST API (API Gateway + Lambda) enabling a Flutter mobile app to create auto-expiring temporary lighting sessions via DynamoDB TTL.
- ESP8266 firmware subscribing to IoT Core topics, driving a 74HC595 shift-register chain via SPI to control up to 16 independent room/lab lights.

Smart Home Automation System

Embedded C · AVR · ADC · PWM · Proteus

- Logic-driven fan speed (PWM) and lighting intensity control based on real-time ADC sensor readings on an AVR microcontroller.
- Interrupt-driven sensor polling for responsive real-time control without blocking the main execution loop.

Door Locker Security System

Embedded C · AVR · UART · I²C · EEPROM

- Dual-MCU security system with password-protected door locking; passwords stored in EEPROM for persistence across power cycles.
- Inter-processor communication over UART and I²C: keypad-driven HMI MCU and a dedicated control MCU managing motor and alarm outputs.

Maze Solving Robot (Logic Simulation)

Digital Logic · 74-series ICs · Multisim

- Wall-following maze-solving algorithm in combinational/sequential hardware logic gates, validated in Multisim before PCB layout.

TECHNICAL SKILLS

Cloud & DevOps	AWS (Lambda, DynamoDB, IoT Core, EventBridge, API Gateway, S3, IAM, CloudWatch), Serverless Architecture, MQTT/TLS, REST APIs
Embedded Systems	AVR, ESP8266, ARM Cortex-M, AUTOSAR, Embedded C
Programming Languages	C, C++, Python, MATLAB, Dart (Flutter)
Networking	CCNA Fundamentals — IP Addressing, Routing, Switching, Subnetting, TCP/IP Stack
Software & Frameworks	Flutter, AUTOSAR, FreeRTOS (basics), Git, VS Code, Eclipse, Code Composer Studio
Hardware & Simulation	Proteus, Multisim, Oscilloscope
Operating Systems	Linux (Administration), Windows

CERTIFICATIONS

Year	Certificate	Issuer
2026	AWS AI Practitioner Course (AIF-C01)	Udemy / Stephane Maarek
2025	AWS Cloud Practitioner Course (CLF-C02)	Udemy / Stephane Maarek
2025	Certificate of Industrial Training	Systel Telecom
2025	Linux Administration Course	Edges for Training
2024	AUTOSAR Software Design Course	Edges for Training
2024	ARM Architecture Course	Edges for Training
2024	Standard Embedded Systems Diploma	Edges for Training